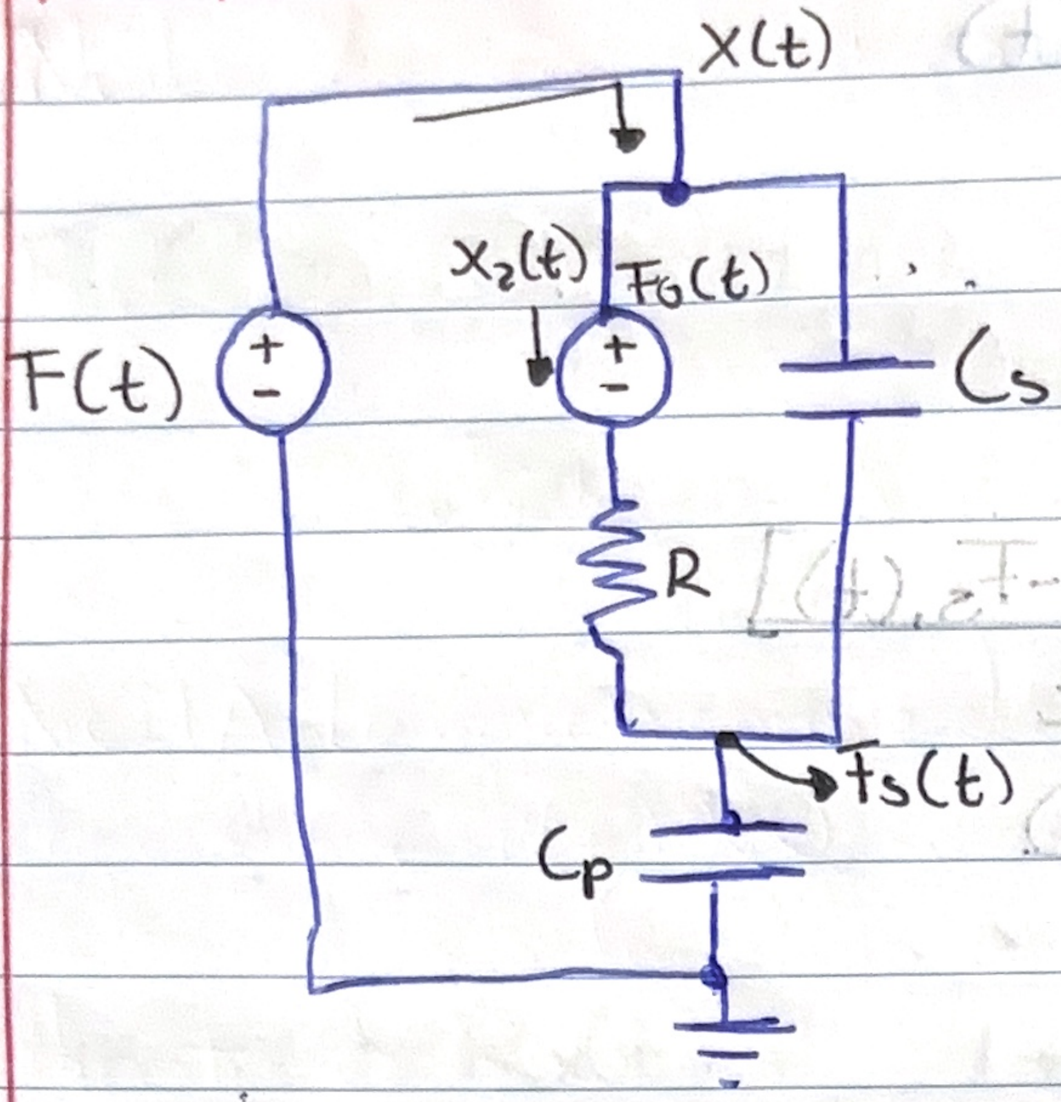
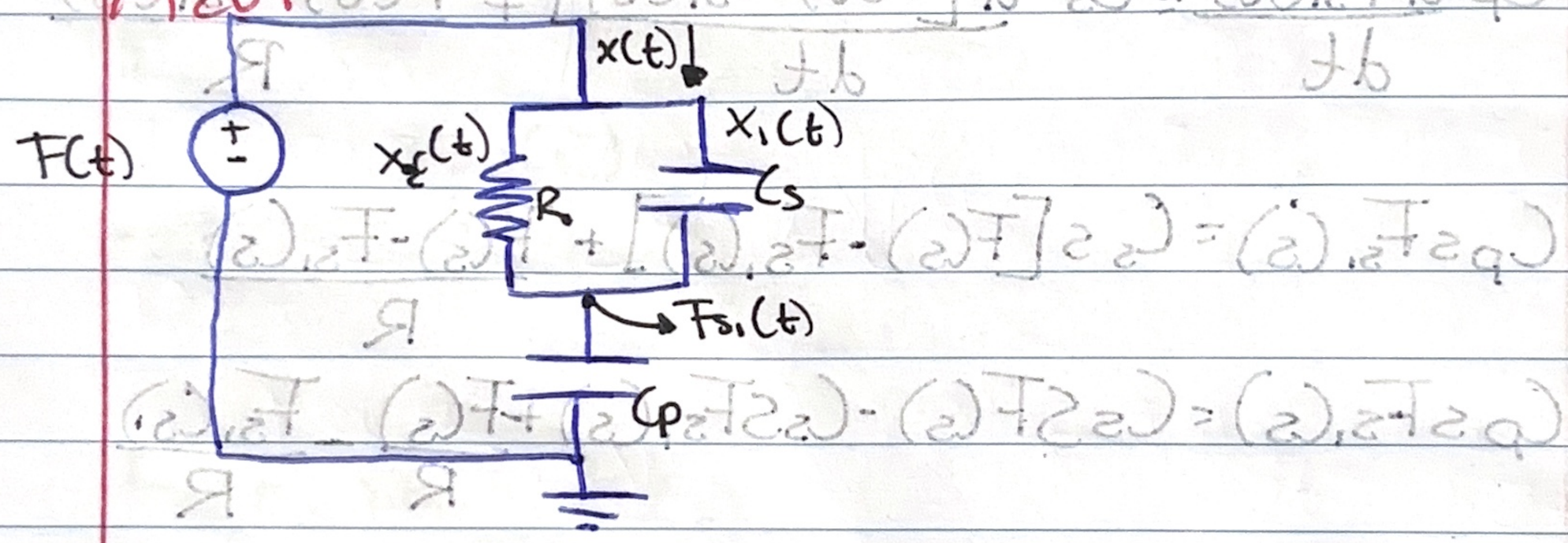
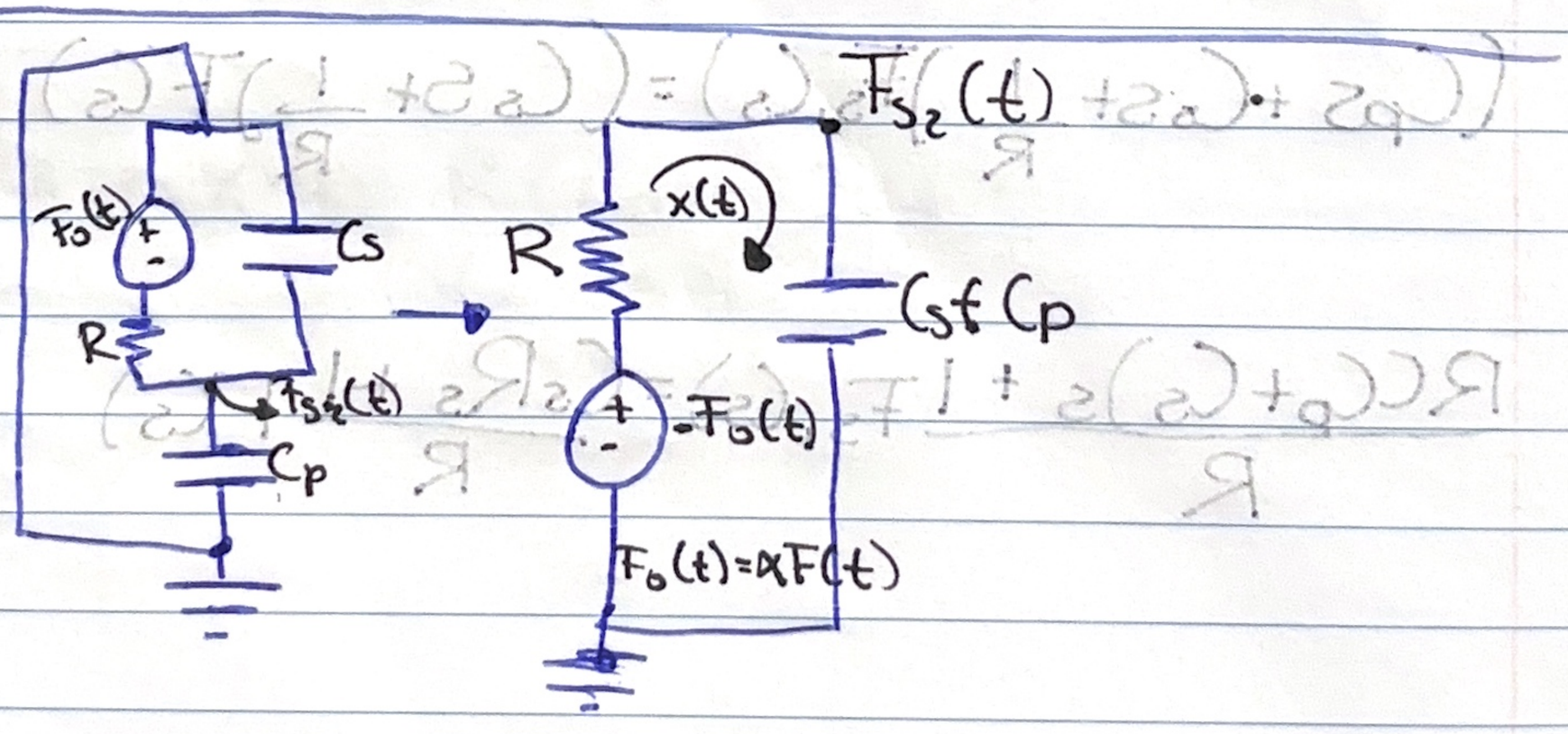
Práctica Musculoesquelética



Azul



Rojo



Analisis de circuito

Circuito Azul ($CF_0(t)$)

$$x(t) = X_1(t) + X_2(t)$$

$$X(t) \; C_p \frac{dF_{s_1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p \frac{dF_{s_1}(t)}{dt} = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p s F_{s_1}(s) = C_s s[F(s) - F_{s_1}(s)] + \frac{F(s) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) = C_s s F(s) - C_s s F_{s_1}(s) + \frac{F(s)}{R} - \frac{F_{s_1}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_S(t) = F_{S_2}(t) + F_{S_1}(t)$$

Funcion de salida 1

$$F_{S_1}(s) = \frac{C_S R_S + 1}{R} F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} \cdot \frac{C_S R_S + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito Rojo $T_-(t)$

$$-\alpha F(t) = R_x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{S_2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)\, s}$$

$$F_{S_2}(s) = \frac{X(s)}{(C_p + C_s)\, s}$$

$$T_-(s) = -\frac{X(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)\, s}$$

$$\frac{F_{S_2}(s)}{F(s)} = \frac{X(s)}{(C_p+C_s)s} = \frac{\alpha}{1+R(C_p+C_s)s+1}$$

$$-\frac{X(s)}{\alpha}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}$$

$$F_s(s) = \frac{C_s R_s + 1 - \alpha}{R(C_p+C_s)s+1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R_s + 1 - \alpha}{R(C_p+C_s)s+1}$$